AITHON 2.0

TITLE PAGE

Problem Statement Track — EdTech

Problem Statement Title — AdaptEd AI - Personalized Learning Assistant.

Theme — Smart Education.

PS Category — Software

Team ID —

Team Name —**SkillMatix.**

IDEA TITLE

PROJECT: PERSONALIZED AI LEARNING ASSISTANT

ONE-LINE TAGLINE: "EMPOWERING CONTINUOUS MASTERY, PERSONALIZED TO YOUR PACE."

PROBLEM



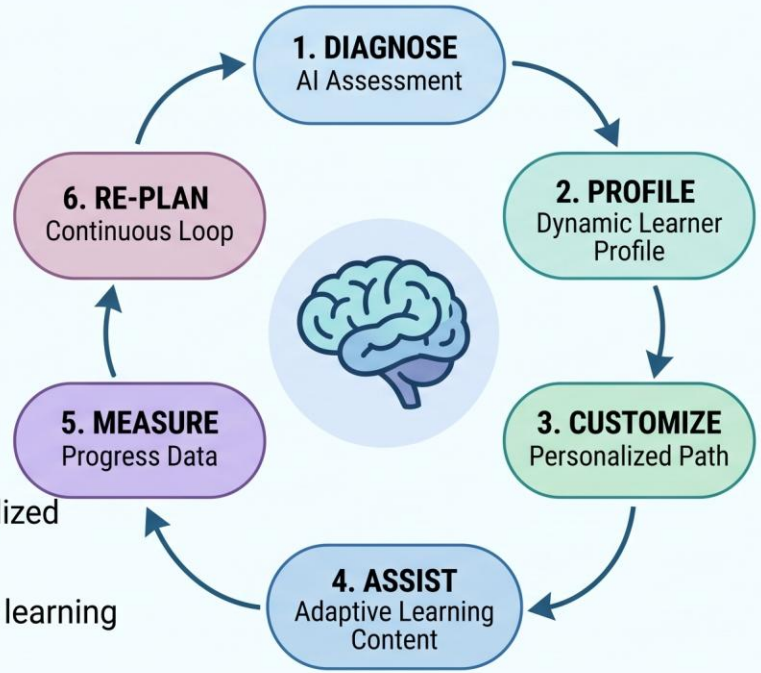
Students struggle with varying knowledge, hidden gaps, and rigid, generic study plans.

PROPOSED SOLUTION

Summary: A continuous adaptive engine providing a dynamic, personalized learning path and content.

Differentiator: Dynamically re-plans learning actions based on real-time data.

An AI-powered adaptive learning platform that continuously personalizes based nisret knowledge, needs, goals.



PERSONALIZED OUTPUTS

- **AI Conversational Assistant**
  - Chat and doubt resolution
- **Dynamic Learning Path**
  - Progress-based goals
- **Adaptive Practice**
  - Practice difficulty matches skill
- **Progress Reports**
  - Skill tracking and milestones

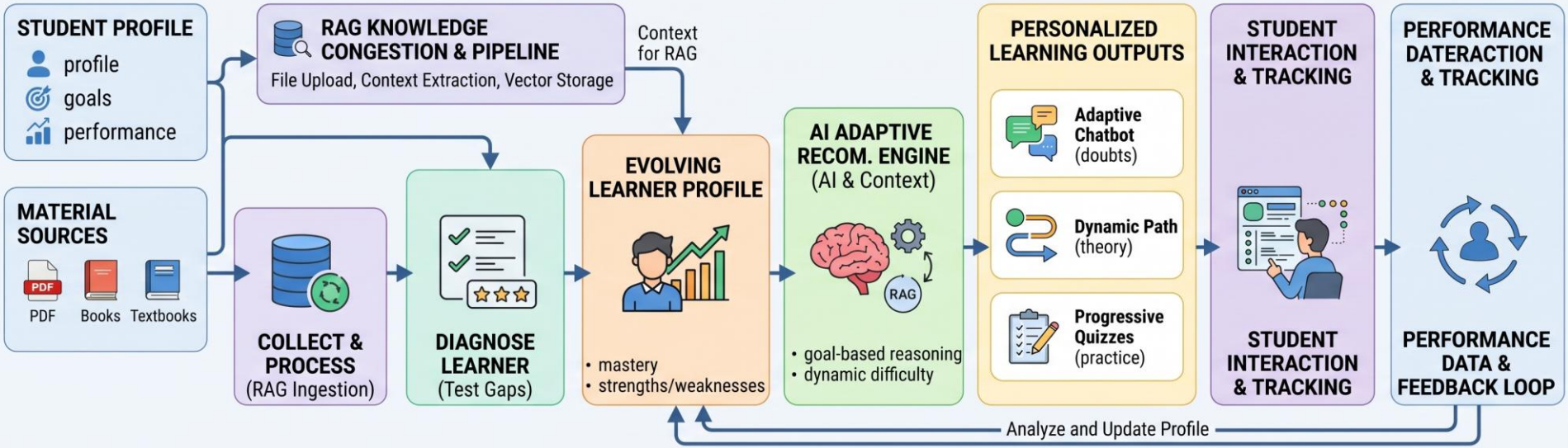
INNOVATION & UNIQUENESS



- Learner-First Focus
- Real-Time Adaptation
- Continuous Cycle

TECHNICAL APPROACH

Full-stack AI + RAG + Adaptive Learning Architecture



TECHNOLOGY STACK

FRONTEND
- Next.js
- React
- TypeScript
- React
- Material UI

BACKEND
- FastAPI
- Python
- Postgres / Supabase

AI & RAG
- Gemini API

RAG Tools:
- PyMuPDF
- Sentence Transformers
- ChromaDB

ANALYTICS & DEV
- Recharts
- Git
- GitHub

FEASIBILITY AND VIABILITY

Practical to build, scalable to deploy, and designed for real student learning needs.

Feasibility

Technical Feasibility
- Uses key mature technologies (e.g., Python, Gemini API, RAG). Modular architecture. (Tech list simplified)

Cost Feasibility
- Low MVP infrastructure needed.
- Uses open-source components and lightweight cloud architecture.

Implementation Feasibility

Frontend → API → RAG Model → Learner Model → Dashboard

Scalability
- **Future Expansion:** Multilingual support, extended subject and student base, institutional deployment, and mobile app.

Challenges & Risks

RAG Accuracy: Content validation required.

AI Hallucination: Quality assurance of outputs.

API Dependency: Latency & limits management.

Student Data Privacy: Protection of personal & performance data.

Limited Initial Data: Impact on initial modeling.

Mitigation Strategies

Step 1
RAG Validation: Better context grounding.

Step 2
Minimal Data Collection: Privacy and security design.

Step 3
Feedback & Evaluation: Quality control and refinement.

Viability

[Server] Low Infrastructure | [Puzzle] Modular | [Chart up] Scalable

IMPACT AND BENEFITS

Transforming one-size-fits-all learning into continuous, personalized learning.



Student & Educator Success

Individualized Paths & Support

- **Personalized Mastery** — Tailored goals & levels
- **Skill Gap Closure** — Targets prerequisite gaps
- **Educator Effectiveness** — Data-driven teaching

Institution & Community Efficiency

Scalable & Effective Learning

- **Scalable Solutions** — Serve more students efficiently
- **Optimized Resources** — Content & faculty time
- **Early Intervention** — Detects at-risk students

Global & Future Impact

Smarter, Inclusive Education

- **Reduced Learning Gaps** — For all backgrounds
- **Equitable Access** — Opportunities for all
- **Future-Ready System** — Inclusive infrastructure

Broad, Sustained Global Impact

- **Reduced Disparities** — Across all groups
- **Universal Quality Access** — For every learner
- **Stronger Educational Ecosystems** — Global readiness

Future Innovations

- Multilingual AI and Accessibility
- Advanced Predictive Learner Models
- Seamless Mobile Access for All

RESEARCH AND REFERENCES

Research-backed design for responsible, *personalized* and *adaptive learning*

RESEARCH FOUNDATION

1. Adaptive Learning
2. Learner Modelling
3. Adaptive Assessment
4. AI-Powered Personalization
5. Responsible GenAI

3. Centtence of arfiemation
4. A onverifevet organizatiion

Focused on individual learners and ethical AI frameworks.

KEY REFERENCES

1. **Adaptive E-learning Review**
   ScienceDirect (2024)
2. **Learner Modelling in Adaptive Systems**
   doi (2024)
3. **AI & Generative AI in Education Guidance**
   UNESCO (2023-2025)

RESEARCH → DESIGN → ADAPT → IMPROVE

CONCLUSION

Conclusion:
Adapted AI integrates learner modelling, RAG, and performance feedback to create a truly individual, continuous student journey.

LEARN → MEASURE → ADAPT → IMPROVE